

HANNAH ARNOLD

First Class BSc (Hons) Artificial Intelligence and Robotics Graduate
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[LinkedIn](#) | [Portfolio](#)

PROFESSIONAL PROFILE

First Class BSc (Hons) Artificial Intelligence and Robotics graduate with practical academic and project experience developing embedded firmware in C using STM32 microcontrollers. Experienced in integrating sensors, motors and peripheral hardware, working with GPIO, timers and I²C communication, and troubleshooting hardware and firmware issues.

Strong interest in embedded firmware development, microcontroller-based systems and hardware-software integration, with additional independent training in embedded C, STM32 development and real-time embedded systems. A problem-solver who enjoys working close to hardware and developing reliable, well-tested software.

TECHNICAL SKILLS

Languages	C, C++, Python, HTML
Hardware & Communication	STM32, GPIO, Timers, I ² C, Sensors, Motors, Circuit Testing
Software Development	Git, GitHub, Visual Studio, Portfolio website
Methods	Algorithm Design, Debugging, Hardware-Software Integration
Professional Strengths	Problem-solving, Organisation, Communication, Teamwork & collaboration, Leadership, Adaptability, Working effectively under pressure

TECHNICAL PROJECTS

Final Year Project: Sensor-Based Embedded System for Roller Coaster Ride Path Analysis | October 2025 - May 2026

Designed and developed an embedded system investigating how roller coaster experiences could be personalised without relying on virtual reality. The system used an STM32 microcontroller to integrate multiple sensors, motors, a camera module and other hardware peripherals for the collection and processing of rider and environmental data.

Developed embedded firmware in C, working with GPIO, timers and I²C communication to interface with connected hardware. The project involved hardware-software integration, electronic circuit testing and systematic debugging of hardware and firmware issues to ensure reliable operation.

Regularly consulted technical datasheets and hardware manuals to understand component specifications, pin configurations, communication requirements and implementation constraints. This required investigating faults across both hardware and software and making changes to improve system reliability and performance.

Personal Technical Portfolio | May 2026 - Present

Portfolio: xxhansterxx.github.io

Developing and maintaining a personal technical portfolio documenting ongoing learning, embedded systems projects and independent technical development. The portfolio includes project documentation, technical experimentation and evidence of continued development beyond university coursework.

EDUCATION

The University Of Staffordshire | September 2023 - May 2026

BSc (Hons) Artificial Intelligence and Robotics - First Class Honours

Relevant areas included embedded systems, robotic programming, artificial intelligence, software development and electronics. Practical work included STM32 microcontrollers, C/C++, Python, GPIO, timers, sensors, I²C communication and hardware-software integration.

City of Stoke-on-Trent Sixth Form College | September 2021 - July 2023

A-Level studies included Mathematics, Physics, Further Mathematics and Extended Project Qualification (EPQ).

ADDITIONAL TRAINING

Embedded Software Development with C Specialization | Educba | August 2026

Completed training focused on embedded C programming, STM32 microcontrollers, software development, hardware interfacing, debugging and real-time embedded systems.

Communication Protocols: UART, USART, I²C, I³C, SPI, GPIO | Udemy | September 2026

Theory based training covering embedded communication protocols including UART/USART, I²C, I³C, SPI, GPIO with protocol signals, packet formats, communication modes, addressing, SPI clock modes and GPIO configuration.

Qt Academy – QML For Beginners | Qt Academy | August 2026

Completed a structured learning path covering QML syntax, Qt Quick, responsive interface development, Model-View-Delegate architecture, animations and transitions. Developed practical experience building and debugging QML applications through hands-on coding challenges, with emphasis on maintainable application development and Qt best practices..

Learn PLC and HMI from Scratch (Basic) Specialization | Packt | August 2026

Completed training in industrial automation covering Siemens S7-1200 PLC programming, TIA Portal, ladder logic, HMI design, system simulation and troubleshooting, with further study of SCL programming, PID control and industrial control systems.

EMPLOYMENT EXPERIENCE

Retail Assistant, Primark | April 2022 - Present

Working in a fast-paced retail environment has developed my customer service, communication and teamwork skills. I am trusted to work across multiple departments and operate till systems accurately, while adapting to changing priorities during busy trading periods.

This role has strengthened my reliability, attention to detail and ability to remain composed under pressure. These transferable skills support my transition into a professional technical environment where collaboration, accountability and consistent performance are essential.

LEADERSHIP AND WIDER EXPERIENCE

Embedded C Study Sessions | April 2026 – May 2026

Organised informal study sessions for Level 5 and Level 6 Robotics students studying embedded C, providing a supportive environment where they could discuss programming problems and technical concepts. I assisted students in troubleshooting coursework and developing their understanding of embedded C, drawing on my previous experience completing the module.

Committee Member and Captain, Macclesfield Warriors Hockey Club | December 2024 - Present

As Team Captain and Committee Member, I contribute to team organisation, communication and fixture coordination. The role requires calm decision-making, leadership under pressure and the ability to support others while maintaining a positive and professional team environment.

Engineering Society, The University Of Staffordshire | September 2023 - May 2024

Participated in practical engineering workshops involving electronics assembly, soldering and prototype development. Activities also included CAD-supported 3D-printing and laser-cutting projects, supporting broader practical engineering awareness.

DRIVING LICENSE

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